Who's the Big Cheese in the University Clubs?

You're new at a university and want to understand the social dynamics of the students. Here's the club membership information:

Club	Members
Drama Club	Sarah, Mike, Emma
Art Club	Emma, Alex
Volunteer Club	Alex, Olivia, James
Sailing Club	Alex, Sophia
Chess Club	Sophia, Ethan, Ava, Noah
Debate Team	Noah, Lily
Math Club	Noah, Lucas
Tennis Club	Noah, Henry

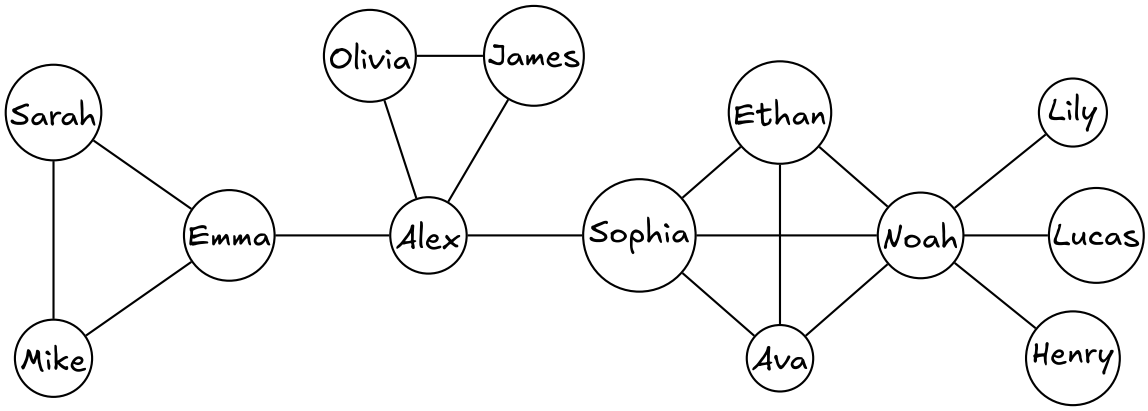
Question 1: Draw a network where students are nodes, and two students are joined by an edge whenever they are in the same club.

Question 2: Without doing any calculations, which student would you approach first if you wanted to spread information quickly about a new inter-club event? Explain your reasoning.

Question 3: Without doing any calculations, which student would you recommend as "Club Coordinator", whose job is to help the different clubs talk to each other? Explain your reasoning.

Keep both answers where you can see them. You will be asked for them again at the end, once you have some numbers.

Here is that network drawn out. Check it against the one you drew — if they differ, find the club you missed.



Question 4: Count the friends. For each student, count how many other students they are joined to. Who has the most?

Student	friends	Student	friends
Sarah		Ethan	
Mike		Ava	
Emma		Noah	
Alex		Lily	
Olivia		Lucas	
James		Henry	
Sophia			

Winner: ____________________

Take the three students who came top in Question 4. They are your candidates for the rest of the sheet. Write their names in the tables below.

Question 5: How far is everyone? For each candidate, count the number of steps along edges to every other student — the shortest route, not just any route. A candidate's distance to themself is 0, so each row has twelve numbers that count: add them and divide by twelve. Who is closest to everybody?

	Sa	Mi	Em	Al	Ol	Ja	So	Et	Av	No	Li	Lu	He	avg

Sa Sarah Mi Mike Em Emma Al Alex Ol Olivia Ja James So Sophia Et Ethan Av Ava No Noah Li Lily Lu Lucas He Henry

Winner: ____________________

Question 6: Who is indispensable? Put your thumb over one candidate and all of their edges. Some pairs of the remaining twelve students can now not reach each other at all. Count those pairs, for each candidate in turn.

Candidate	sizes of the groups left behind	pairs cut off

Mechanics, not the answer: if covering someone leaves three groups of sizes 2, 3 and 4, then the pairs cut off number $2 \times 3 + 2 \times 4 + 3 \times 4 = 26$.

Winner: ____________________

Question 7: Pass the scores around. Give every one of the thirteen students one point. Now play this game: each student throws away their own score and takes instead the sum of their friends' scores. Play it twice.

Round 1 is quick — you already did it in Question 4, so write that column in. For round 2, add up the number of friends of each of your candidate's friends.

Candidate	after round 1	after round 2

Winner: ____________________

Question 8: The point of the whole sheet. Copy your four winners into the four boxes.

most friends (Q4)	
shortest average distance (Q5)	
most pairs cut off (Q6)	
highest score after two rounds (Q7)	

How many different names did you write? ________

Nothing about the network changed between those four boxes. Only the question did. Write one sentence for each box saying what question it was really asking about a student.

Question 9: Now name a situation — a real one, at a real university — in which each of your winners would be the right person to pick, and the others would be the wrong choice.

Question 10: Go back to Questions 2 and 3. With the numbers in front of you, do you still give the same two names? Which of your four measures matches the question "who spreads news fastest?", and which matches "who should be Club Coordinator?"

Question 11: Break it. A Robotics Club is formed, and Noah and Emma both join it. Add that one edge to the drawing on page 3.

Redo Question 6 in your head for each candidate. One of them goes from being indispensable to being dispensable — who, and why? Does that person still lie on plenty of the shortest routes, even so?

Question 12: In Question 7 you played two rounds of passing scores. You don't need to do the calculation — just describe the recipe you would hand to a computer so that it could keep playing that game forever, on a network of a million students. What happens to the size of the numbers as the rounds go by, and what would you have to do about it?